Unit 1 Parent Functions & Transformations Study Guide

Objectives…Remember that big colored sheet of paper you taped into your notebook at the beginning of the unit??

A. Parent Functions

- ☐ I can identify the graph, name, and equation of the following parent functions:
  - ☐ Linear, Absolute value, Quadratic, Cubic, Rational, Square Root, Cube Root, Exponential, Logarithmic

B. Transformations of Functions

- ☐ I can translate functions horizontally and vertically.
- ☐ I can reflect functions over the x-axis and y-axis.
- ☐ I can vertically dilate functions.
- ☐ I can describe transformations of functions from a graph or equation.

C. Analyzing Graphs of Functions

- ☐ Using a graph, I can identify the following key features:
  - ☐ x-intercept(s) & y-intercept
  - ☐ Domain & Range
  - ☐ Increasing & Decreasing Intervals
  - ☐ Positive & Negative Interval(s)
  - ☐ Maximum & Minimum (Extrema)

Essential Questions

1. What are parent functions?
2. What are the types of basic parent functions?
3. What are the characteristics of the basic parent functions?
4. What does it mean to transform a function?
5. How can a function be transformed?
6. What are the key features of graphs?

Topics Covered

A. Lesson 1 - Parent Functions

- Name the basic parent functions
- Identify graphs of basic parent functions
- Write the equations of basic parent functions

B. Lesson 2 - Transformations of Functions

- Transform parent functions (translations, reflections, vertical stretches/shrinks)
- Transform functions (translations, reflections, vertical stretches/shrinks

C. Lesson 3 - Analyzing Graphs of Functions

- Identify intercepts, domain & range, increasing & decreasing intervals, positive & negative intervals, and maximums & minimums of graphs
- Use correct notation when analyzing graphs & identifying characteristics

How can I practice/study for this assessment?

Your notes, practice, and homework creates a "practice guide"!

- ☐ Complete any unfinished practice problems
- ☐ Redo practice problems
- ☐ DeltaMath
- ☐ Visit our Google Classroom page for digital versions of in class activities!!
- ☐ Visit our Google Classroom page for any additional digital versions of practice!!